Semiconductor basics: controlling electricity

How diodes, transistors and MOSFETs work | SEMICONDUCTOR BASICS

Smartphones, cars, factories and AI data centers do not run on computing chips alone. They need components that condition current and voltage, amplify signals and switch electricity.

01 Diode

A component that conducts current in one direction and makes reverse flow difficult. Main roles: rectification, reverse-current prevention and circuit protection. Typical uses: power supplies, chargers and automotive circuits.

02 Transistor

A general term for components that control current using electrical signals. Main roles: signal amplification and switching current on/off. Typical uses: CPUs, communications circuits, sensors and control circuits.

03 MOSFET

A type of transistor that controls current with voltage applied to a gate. Main roles: switching, signal processing and power conversion. Typical uses: integrated circuits, server power supplies, automotive power supplies and inverters.

Classification point

MOSFETs are included within transistors. These are not three separate classifications at the same level.

How do the three components work?

Conduct in one direction; control with a signal; switch with voltage

Diode | A check valve for current

Used for “rectification,” extracting current that flows in one direction from alternating current. Rectification alone leaves ripple in the output; a stable DC supply also needs smoothing and voltage regulation.

AC → rectification → smoothing and voltage regulation → stable DC

Transistor | Amplification and switching

A small input signal controls the output current. Analog circuits amplify signals; digital circuits use on/off states for logic operations. The energy needed for amplification comes from the power supply.

MOSFET | Gate voltage controls current

A MOSFET is a type of field-effect transistor. Gate voltage controls current between source and drain. Steady-state gate current is small, but switching requires charging and discharging the gate.

From tiny logic devices to power conversion

Some MOSFETs process signals inside integrated circuits; power MOSFETs handle large amounts of power. Speed and losses depend on material, voltage rating, structure and drive conditions.

Where are they needed?

Their shared role is to convert and control electricity into a form suited to its use

AI data centers | Fine voltage conversion

Inside servers, they produce the low-voltage, high-current supplies needed by GPUs and other devices. MOSFETs and power-management ICs handle power conversion; reducing conversion losses lowers heat and power consumption.

EVs and hybrids | Electricity drives motors

Inverters convert battery DC into AC suited to motor operation. Power semiconductors are also used in chargers and DC-DC converters. SiC MOSFETs are used in high-voltage power conversion and other applications.

Solar and batteries | Connecting to the grid

Power conditioners and other equipment convert DC from solar panels or storage batteries into AC and adjust voltage and frequency. Semiconductor control is essential to produce electricity suited to the application.

Factories and robots | Precise motion control

Inverters and servo drives control motor speed and torque. Power semiconductors supply the required power appropriately.

Material differences

Si is silicon; SiC is silicon carbide; GaN is gallium nitride. Their properties differ, so materials are selected according to the application.

How should supply and demand be assessed?

Check product-specific conditions rather than treating all “chip shortages” as one

Even MOSFETs are not always interchangeable

Voltage rating, current capability, losses, heat dissipation and packaging differ. One type may be in short supply while another has spare capacity. Demand growth alone does not establish a universal shortage.

More production needs equipment and manufacturing expertise

Power semiconductors also use production equipment for 200 mm wafers and other sizes. Wafer diameter and process node, which indicates circuit miniaturization, are separate measures. Supply depends on process and equipment capacity for each product.

For SiC, material quality and yield matter too

Crystal and wafer quality, processing technology and manufacturing yield affect SiC capacity and cost. Even after expanding factories, process improvements are needed to achieve stable mass production.

Automotive products take time to evaluate and qualify

Automotive components require high reliability. Even similar products cannot necessarily be substituted without evaluation or design changes.

Condition, amplify and switch electricity: these basic functions support computing, communications and electrification.